

703159 VU

Data-driven Analysis of Geodata

Session #03: Some probability theory and statistics

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CONTENT

- Statistical thinking
- Notions of probability and population
- Conditional probability, independence, conditional independence
- Information-theoretic measures [ipynb](#)
- Spurious correlations in non-stationary time series [ipynb](#)
- Primer on selected probability distributions [ipynb](#)

ipynb in Jupyter notebook.

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Conditional probability & Bayes' thm.

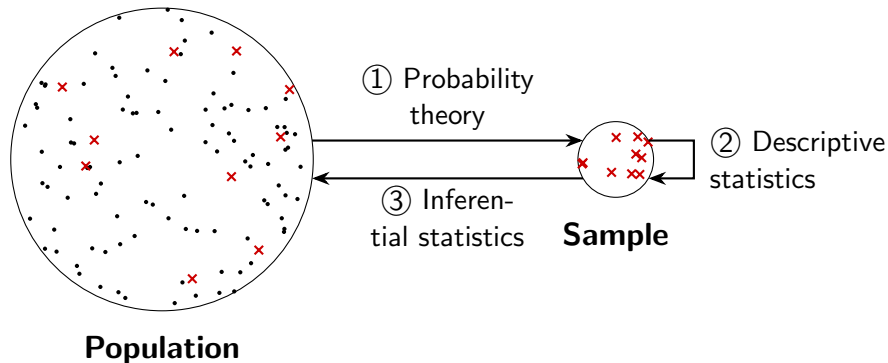
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Statistical thinking



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The statistical workflow

Central dogma of statistics:

The analysis of a small random sample enables drawing inferences about the entire population.

- ① **Probability theory.** Population $\rightarrow$ Sample.
Given (known or assumed) properties of the population, what should our sample look like?
- ② **Descriptive statistics.** Sample $\rightarrow$ Summarise and describe the observed sample as-is.
- ③ **Statistical inference.** Sample $\rightarrow$ Population.
Given the observed sample, what can we infer about the underlying population?

① and ③ are inverses: Inference is probability reasoned backwards.

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Interpretations of probability

Interpretations of probability

- as propensity, a property of a "random system"
 - as long-run frequency of outcomes
 - as degree of belief/state of knowledge (evidential probability)
- Propensity: a physical, inherent disposition or tendency of a specific setup (physical system) to produce certain outcomes. Applicable to single-case outcomes of random processes.
 - Frequentist probability: Observed relative frequency of the occurrence of an event in an infinite number of repeated trials. Not applicable to single outcomes.
 - Bayesian probability: An agent's subjective degree of belief or state of knowledge. Applicable to any statement, even when no random process is involved.

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Where does probability live — in the world or in the mind?

- What can have probability?
- Where does probability live?

Frequentist interpretation	Bayesian interpretation
Probability lives in the “world out there”, in the data-generating process.	Probability lives in the mind of the observer.
Describes the long-run relative frequency of outcomes of repeated random experiments (aleatoric uncertainty).	Describes the agent’s state of knowledge of any proposition (even when no randomness is involved: epistemic uncertainty).
Aleatoric inherent randomness in the data; parameters fixed (albeit unknown).	Epistemic uncertainty in the parameters; data are fixed once observed ¹ .

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¹ Latin root of the word *data*: “that which is given”.

Types of Population

- **Literal population.** Classical statistics infers from a sample to a real, countable group.
- **Virtual population.** Repeated measurements vary slightly due to finite precision, conceptualised as draws from a hidden theoretical distribution.
- **ML/big data context.** Often *all* possible data are available: no sampling involved, virtually error-free.

EXAMPLE U.S. exit polls: $< 50\,000$ out of 160M voters ($< 0.05\%$) polled.

EXAMPLE Instrument noise in a lab measurement.

EXAMPLE Amazon's transaction database *is* the population; there is no larger one.

So what does probability mean when you have all the data?

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Types of Populations

Literal population	Virtual population	Metaphorical population
A real, identifiable set of countable units from which a sample is literally drawn.	The abstract set of all values obtainable by repeating the same measurement procedure under identical conditions.	An imaginary set of alternative realisations if the same data-generating situation was repeated.
Uncertainty from sampling variability (what was selected) and coverage gaps (what was missed).	Uncertainty from finite instrument precision and measurement noise.	Uncertainty is epistemic: we <i>choose</i> to act as if the observed data is one draw from an <i>imagined</i> space of compatible possibilities.
Counting birds in forest: Visible birds $\subset$ all birds.	One realized measurement $\subset$ ensemble of measurements.	Actual realisation $\subset$ multiple compatible realisations

[Spiegelhalter, 2020, p.91ff]

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Probability: The formal mathematical expression of uncertainty.

Let $P(A)$ be the probability for an event A . Then the rules of probability are:

Bounds: The probability of any event A must fall between 0 and 1 (inclusive):

 $0 \leq P(A) \leq 1$, with

$$P(A) = \begin{cases} 0, & \text{if } A \text{ is impossible (necessarily false)} \\ 1, & \text{if } A \text{ is certain (necessarily true)} \end{cases} \quad (1)$$

Total probability rule (sum rule): The sum of the probabilities of all possible outcomes ω in a sample space Ω must equal 1,

$$\sum_{\omega \in \Omega} P(\omega) = P(\Omega) = 1. \quad (2)$$

[Weisberg, 2018; Skiena, 2017; Deisenroth et al., 2020]

Addition rule, union, and exclusivity

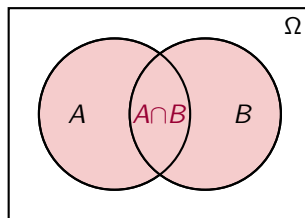
Addition rule. The union $A \cup B$ (either A or B , or both):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (3)$$

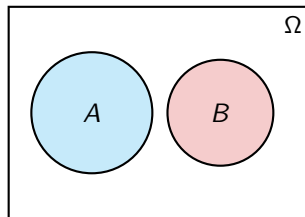
Subtracting $P(A \cap B)$ corrects for double-counting the overlap — the inclusion-exclusion principle.

If A and B are **mutually exclusive** ($A \cap B = \emptyset$, they cannot both occur), then $P(A \cap B) = 0$ and the rule simplifies:

$$P(A \cup B) = P(A) + P(B) \quad [A \cap B = \emptyset] \quad (4)$$



Union $A \cup B$



Mutually exclusive, disjoint

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Multiplication rule, intersection, and independence

Multiplication rule. The intersection $A \cap B$ (both A and B occur):

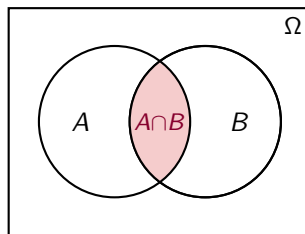
$$P(A \cap B) = P(A) \cdot P(B \mid A) \quad (5)$$

If A and B are (marginally) **independent**, knowing one tells you nothing about the other, then $P(B \mid A) = P(B)$. The rule simplifies to:

$$P(A \cap B) = P(A) \cdot P(B) \quad [A \perp\!\!\!\perp B] \quad (6)$$

Equivalently: $A \perp\!\!\!\perp B$ iff $P(B \mid A) = P(B)$.

Mutually exclusive events are not independent!
(unless one has zero probability).



Intersection $A \cap B$

NOTATION $P(A \cap B) \equiv P(A, B)$:
same thing, two conventions.

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Conditional Probability

Probability of event B given that event A already happened, notation: $\mathcal{P}(B | A)$.

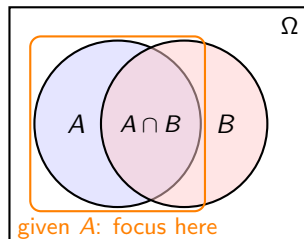
Key idea: Restrict the sample space Ω to outcomes where A has occurred, then ask how probable B is within that restricted space.

From the product rule:

$$P(A, B) = P(B | A) \cdot P(A) \quad (7)$$

$$\Rightarrow P(B | A) = \frac{P(A, B)}{P(A)} \quad (8)$$

EXAMPLE “Probability of rain given that it’s already cloudy”: $P(\text{rain} | \text{cloudy})$: how often does it rain *and* look cloudy, relative to how often it’s cloudy at all?



$$P(B | A) = \frac{\text{area of } A \cap B}{\text{area of } A}$$

Note: $P(A, B) \equiv P(A \cap B)$, same thing, different notation.

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Conditional probability: Order matters

Conditioning is not commutative:

$$P(B | A) \neq P(A | B) \quad \text{in general} \quad (9)$$

EXAMPLE All popes are Catholic, but only one Catholic is the Pope:

$$P(\text{Catholic} | \text{Pope}) = 1 \quad P(\text{Pope} | \text{Catholic}) \approx 0$$

Why it matters. Confusing $P(B | A)$ with $P(A | B)$ is one of the most common reasoning errors in statistics — known as the *prosecutor's fallacy*.

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Bayes' Theorem

RECALL: $P(A, B) = P(B, A)$ (joint prob. is symmetric), so:

$$P(B | A) P(A) = P(A | B) P(B)$$

Dividing both sides by $P(A)$ gives **Bayes' theorem**:

$$P(B | A) = \frac{P(A | B) P(B)}{P(A)}$$

(10)

The denominator $P(A)$ is often obtained by marginalising out B :

$$P(A) = \sum_b P(A, b) = \sum_b P(A | B=b) P(B=b)$$

This is the computationally hard part.

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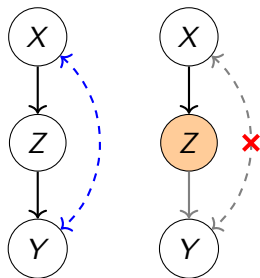
Causal chain and Conditional independence

- Consider the chain $X \rightarrow Z \rightarrow Y$. Since X influences the mediator Z and Z influences Y , information flows from X to Y : they are generally **marginally dependent**.
- Conditioning on Z (fixing it to a known value) **blocks** the path. Changing X no longer affects Y .
- Therefore: $X \perp\!\!\!\perp Y \mid Z$.

EXAMPLE Fire $\rightarrow$ smoke $\rightarrow$ Alarm.

Not knowing smoke: an alarm suggests fire.

Knowing smoke: the alarm adds nothing: fire is already explained by smoke, and the alarm is only caused by smoke, not the fire.



Orange = conditioned (fixed) node.

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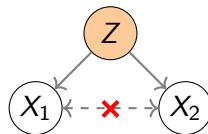
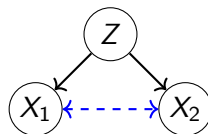
Forks and confounders

- Consider the fork $X_1 \leftarrow Z \rightarrow X_2$. A common cause or confounder Z induces a (spurious) correlation between X_1 and X_2 : they are **marginally dependent**.
- Conditioning on ('filtering') Z removes this shared cause. With Z fixed, X_1 and X_2 no longer co-vary: the path is **blocked**.
- Therefore: $X_1 \perp\!\!\!\perp X_2 \mid Z$.

EXAMPLE Ice cream sales $\leftarrow$ temperature $\rightarrow$ Drownings.

Without conditioning: both rise together in warm weather: a spurious correlation.

Conditioning on temperature: comparing only days of equal temperature, the correlation vanishes.



Orange = conditioned (fixed) node.

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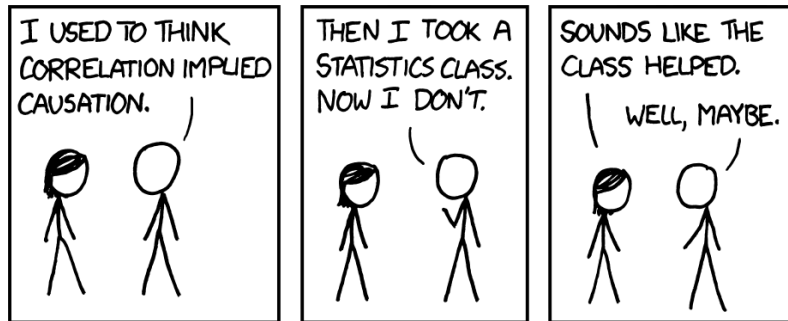
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Dependence does not imply causation



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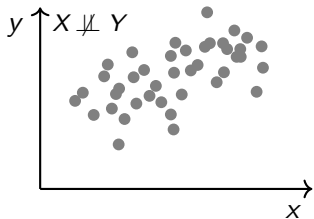
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Hidden clusters, spurious correlation, conditional independence



Simpson's paradox

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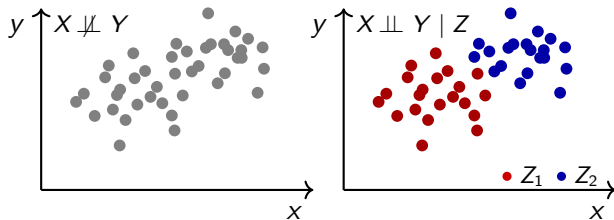
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Simpson's paradox

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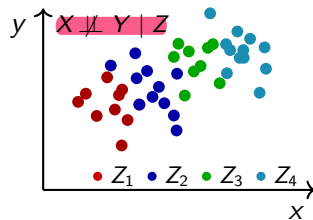
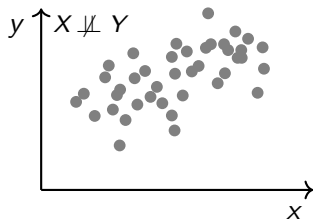
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Statistical association $\neq$ Causal relationship

A statistical **association** is a relationship between two or more variables where changes in one variable tend to coincide with changes in another, indicating they are not statistically independent. It suggests variables are linked, but it does not imply that one causes the other ('correlation does not equal causation').

- **Association** enables learning: the degree to which some of the variability of one variable can be accounted for by the other.
- Correlation is a special case of *linear* association.
- Association $\neq$ Causation: Association simply means a statistical dependency exists, not (necessarily) that one variable dictates the other.

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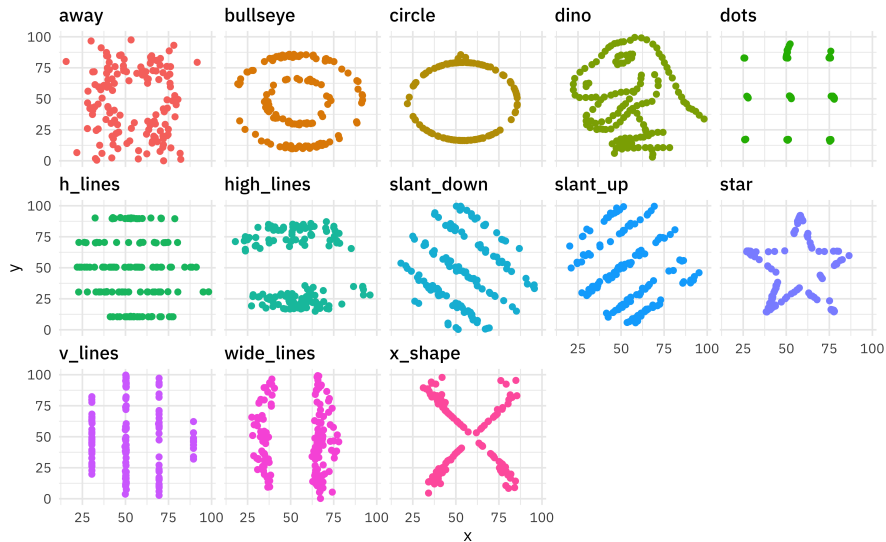
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Choosing a probability distribution

Discrete or continuous? • Symmetric or asymmetric? • Heavy tails, outliers?

Support: $[-\infty, +\infty]$, $[0, +\infty]$, $[0, 1]$?

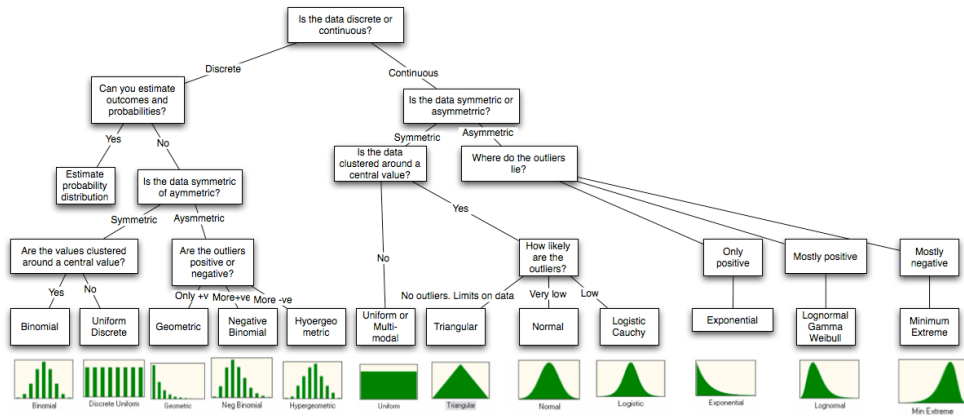


Chart by Aswath Damodaran (NYU), <https://pages.stern.nyu.edu/~adamodar/pdfiles/papers/statprimer.pdf>

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